

message on the screen along with musical sound produced from the internal speaker. You will hear the major scale of do, re, me, fa, so, la, ti, do' (whose Indian version is sa, re, ga, ma, pa, dha,

ni, sa') musical tunes in the internal speaker. The program stops and exits the application by itself after producing these tunes. The screenshot of the program output is shown in Fig. 1.

The musical sound of this program output is not of standard pitch but shows how the available 'C' functions can be used to generate sound from the internal speaker.

The program uses two main functions: sound() and delay(). It starts with the stdio.h header file, which contains the basic input and output functions. 'dos.h' is a special header file required by the sound function. 'my_sounds()' tells the compiler that this is a new user-defined function. 'main()' is the main function of the 'C' program and all the user-defined functions are called from this function. The clrscr() function clears the previous screen whenever the program is run.

The printf() function prints the text on the screen.

The sound() function receives the frequency as the input parameter and plays it until the nosound() function is called. For example, 'sound(500)' produces a beep sound at a frequency of 500 Hz. 'delay(1000)' allows the beep to continue for 1000 milliseconds. The nosound() function turns off the sound in the speaker.

The other program, beep2.cpp, is developed using Microsoft Visual C++. When you run this program, you will hear do, re, me, fa, so, la, ti, do', do', ti, la, so, fa, me,

re, do. The screen displays the message 'Hello, this is how your PC works!' in the first line and 'Press enter key to exit the program' in the second line, until you hit 'Enter' key from the keyboard. The screenshot of the program output is shown in Fig. 2.

The program starts with the ios-stream header file, which contains input and output definitions.

'windows.h' is a Windows-specific header file that contains declarations for all the functions in the Windows API and all the data types used by the various functions and sub-systems.

The 'using namespace std;' line tells the compiler to use a group of functions that are part of the standard library (std). Namespace 'std' contains all the classes, objects and functions of the standard C++ library. The semicolon is part of the syntax of C and C++. In C++, the cout object is used to display text. It uses '<<' symbols, known as insertion operators, to indicate what to output. The double quotes tell the compiler that you want to output the literal string/text on the output screen as entered by you between these quotes.

The most important function in this program is 'beep()'. This function generates a simple tone from the PC's speaker. The syntax is beep(frequency,duration). For example, the beep(523.2500,500) code generates a beep at 523.2500 Hz for a duration of 500 milliseconds.

The last important function is 'cin.get()'. This is another call function. It gets an input and expects the user to hit Enter key from the keyboard. Many compiler environments will open a new console window, run the program and then close the window. This command keeps that window from closing until you hit 'Enter' key. The cin.get() function gives you the time to see the program run and text appear on the screen.

EFY note. The source codes are included in this month's EFY-CD and are also available on www.efymag.com website.

Musical Notes and Their Frequencies

Note	Frequency (Hz)	Note	Frequency (Hz)
Octave 0		F# / Gb	184.997
C	16.351	G	195.998
C# / Db	17.324	G# / Ab	207.652
D	18.354	A	220
D# / Eb	19.445	A# / Bb	233.082
E	20.601	B	246.942
F	21.827		
F# / Gb	23.124	Octave 4	
G	24.499	C	261.626
G# / Ab	25.956	C# / Db	277.183
A	27.5	D	293.665
A# / Bb	29.135	D# / Eb	311.127
B	30.868	E	329.628
		F	349.228
Octave 1		F# / Gb	369.994
C	32.703	G	391.995
C# / Db	34.648	G# / Ab	415.305
D	36.708	A	440
D# / Eb	38.891	A# / Bb	466.164
E	41.203	B	493.883
F	43.654		
F# / Gb	46.249	Octave 5	
G	48.999	C	523.251
G# / Ab	51.913	C# / Db	554.365
A	55	D	587.33
A# / Bb	58.27	D# / Eb	622.254
B	61.735	E	659.255
		F	698.456
		F# / Gb	739.989
Octave 2		G	783.991
C	65.406	G# / Ab	830.609
C# / Db	69.296	A	880
D	73.416	A# / Bb	932.328
D# / Eb	77.782	B	987.767
E	82.407		
F	87.307	Octave 6	
F# / Gb	92.499	C	1046.502
G	97.999	C# / Db	1108.731
G# / Ab	103.826	D	1174.659
A	110	D# / Eb	1244.508
A# / Bb	116.541	E	1318.51
B	123.471	F	1396.913
		F# / Gb	1479.978
Octave 3		G	1567.982
C	130.813	G# / Ab	1661.219
C# / Db	138.591	A	1760
D	146.832	A# / Bb	1864.655
D# / Eb	155.563	B	1975.533
E	164.814		
F	174.614		